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Probabilistic spatio-temporal machine learning methods for distribution system state forecasting ⋆

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ABSTRACT

This paper investigates probabilistic, data-driven methods for short- to mid-term distribution system state forecasting to support proactive grid operation. We compare multiple forecasting methods that integrate spatial and temporal learning, ranging from baseline models based on standard load profiles to fully integrated spatio-temporal neural networks. In particular, we evaluate the A3T-GCN architecture, which combines graph convolutions, gated recurrent units, and temporal attention for multi-horizon state prediction. Using both synthetic SimBench and real low-voltage grids, we assess deterministic and probabilistic forecast accuracy as well as the capability to anticipate critical operating conditions. Our results show that purely temporal models already achieve competitive forecasting performance, capturing a large share of system dynamics. Explicitly modeling spatial–temporal dependencies improves robustness across networks. Among the evaluated methods, the A3T-GCN achieves the most stable performance across grids and forecasting horizons, demonstrating its potential as a transferable framework for reliable, uncertainty-aware state forecasting in future distribution systems.

1. Introduction

The global energy system is transitioning from centralized generation to highly decentralized distribution networks with large shares of renewable and flexible resources. Distributed Energy Resources (DERs) such as photovoltaic (PV) systems, electric vehicles (EVs), battery storage, and heat pumps are increasingly connected at the grid edge. While these technologies enable deep decarbonization, they also introduce significant operational challenges for Distribution System Operators (DSOs), as local demand and generation become more volatile, uncertain, and spatially heterogeneous [1].

To operate such systems securely and efficiently, DSOs increasingly require short- to mid-term situational awareness beyond the current system state. Forecasts of future system states enable proactive operation, supporting applications such as preventive congestion management, voltage control, and network reconfiguration. Unlike traditional Distribution System State Estimation (DSSE), which reconstructs the current grid state from measurements, *state forecasting* aims to predict future electrical states over multiple time horizons. This task is particularly challenging in low-voltage distribution networks, where sparse sensing, limited observability, and uncertain topology coincide with highly dynamic DER-driven behavior [2,3]. Classical physics-based forecasting pipelines typically rely on separate load and generation forecasts combined with numerical power-flow simulations, which can be sensitive to forecast errors and do not explicitly account for uncertainty.

Recent advances in machine learning (ML) provide new opportunities to address these challenges. Temporal models such as recurrent neural networks and self-attention-based Transformers excel at learning complex time dependencies from historical data, while Graph Neural Networks (GNNs) naturally encode the spatial structure of electrical networks. Combining temporal and spatial learning promises data-driven state forecasting methods that are both accurate and scalable, even under high DER penetration and partial observability.

Against this background, this paper investigates data-driven approaches for probabilistic distribution system state forecasting with integrated spatial and temporal modeling and provides a systematic comparison of modeling strategies with respect to predictive performance and their ability to anticipate operationally critical grid states.

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2. Literature review

Traditional DSSE methods infer nodal voltages and phase angles from limited and noisy measurements using optimization-based formulations of the power-flow equations, often augmented with pseudo-measurements to mitigate sparse sensing [4]. While physically interpretable, these methods can be computationally demanding, sensitive to modeling errors, and prone to convergence issues in real-time or large-scale settings, particularly in low-voltage networks [5]. Moreover, DSSE focuses on reconstructing the present state and does not directly address the need for forecasting future operating conditions.

Data-driven approaches have gained traction as an alternative, leveraging historical measurements to learn direct mappings from observations to system states. In particular, Graph Neural Networks (GNNs) have emerged as a powerful tool by embedding the electrical topology into the learning process. GNN-based models can capture spatial correlations between nodes and branches and have demonstrated competitive accuracy and scalability compared to classical methods, especially when combined with physics-aware constraints or hybrid formulations [6,7]. Standardized benchmarks such as the Distribution System Machine Learning (DSML) framework further facilitate systematic evaluation across network types and operating conditions [8].

However, most DSSE approaches process time steps independently, limiting their predictive capability. Integrating load forecasts partially addresses this [9,10], but temporal dependencies are typically not learned jointly with spatial structure. Recent efforts to include spatial modeling, including graph-assisted load forecasting, improve robustness and short-term predictions [11], yet do not perform true state forecasting over longer horizons.

Beyond static or step-wise formulations, a growing body of work explores end-to-end architectures that map historical measurements directly to future system states or violation indicators using recurrent, convolutional, or attention-based models. By learning temporal patterns from data, these approaches capture complex dynamics without explicit physical modeling and have shown strong performance in specific scenarios [12–14]. Classical time-series models are largely agnostic to spatial structure: they primarily model temporal dependencies and treat measurement streams independently or rely on implicitly shared representations. While this facilitates deployment across systems, network topology is not explicitly embedded. Consequently, spatial dependencies are only learned implicitly.

Recognizing the complementary strengths of topology-aware and sequence-based models, recent research has moved toward explicitly *spatio-temporal* learning frameworks that integrate graph representations of the grid with temporal sequence modeling. In these approaches, GNN layers encode electrical connectivity while recurrent or attention-based modules capture temporal evolution. In power systems, applications of these frameworks include PV and wind forecasting [15–19] as well as short-term load prediction [20,21]. Across these tasks, jointly modeling spatial and temporal dependencies has consistently improved short-term predictive accuracy compared to models that treat these dimensions separately. Architecturally, many of these designs are inspired by advances in related domains, most notably traffic flow prediction, where tightly coupled graph–temporal networks have achieved state-of-the-art performance [22]. Both traffic and power systems can be represented as sparse, sensor-instrumented networks with fixed topology, localized interactions along edges, and strongly correlated multivariate time series driven by exogenous factors (e.g., weather, demand patterns). While the underlying principle of combining localized spatial interactions with long-range temporal dependencies is transferable, adapting such architectures to power systems is nontrivial. Electrical networks are governed by physical constraints, exhibit different node semantics, and are characterized by rare but operationally critical events. Nevertheless, initial studies adapting or transferring these hybrid architectures to power system data report promising results [23–25]. However, these studies typically focus on deterministic predictions, short horizons, or specific use cases, leaving questions of probabilistic forecasting and operational relevance largely open.

Finally, while standard accuracy metrics such as MAE or RMSE are widely reported, they do not fully capture the value of forecasts for grid operation. From a DSO perspective, forecasts are most useful when they enable preventive actions to avoid voltage or thermal violations. Some studies address this by combining ML-based injection forecasts with power-flow simulations [26–28], but forecast uncertainty is often treated implicitly or outside the learning loop.

Despite progress, two gaps remain: First, spatio-temporal models for state forecasting over mid-term horizons are underexplored, and second, evaluations often ignore operational metrics reflecting the ability to prevent constraint violations. This work addresses these gaps by proposing a probabilistic, data-driven method that integrates temporal sequence modeling with graph-based grid representations, forecasting system states for up to 48 h and assessing both numerical accuracy and the detection of critical conditions such as voltage limit violations and line overloads.

Based on these gaps, this paper makes the following contributions:

- We investigate various combinations of spatial and temporal models for probabilistic distribution system state forecasting, starting from naive approaches based on standard load profiles and numerical power flow computation and ending at fully integrated spatio-temporal neural networks.
- We systematically evaluate the proposed methods across multiple realistic distribution networks, accounting for varying DER penetration and sensor coverage based on forecasting performance.
- Based on the developed forecasting methods and our evaluation framework, we assess the capability of anticipating critical grid states including voltage violations and line overloads.

3. Methods

3.1. Problem statement

Let a distribution network be represented by a set of buses $\mathcal{N}$ with $|\mathcal{N}| = n$ and a set of lines $\mathcal{E}$. We denote the complex voltage at bus i by $V_i = |V_i|e^{j\phi_i}$. The full system state is

$$\mathbf{x} = \left(|V_1|, \dots, |V_n|,\ \phi_1, \dots, \phi_n\right)^\top, \tag{1}$$

with a reference bus angle fixed, e.g. $\phi_{\text{slack}} = 0$. The objective of *state forecasting* is: given historical measurements $\mathbf{z}_{t-T_p+1:t}$ including past time steps T_p up to time t and exogenous covariates $\mathbf{ex}_t$, predict the future state $\mathbf{x}_{t+\tau}$ for multiple time steps up to the horizon T_f: $\tau \le T_f$. We also denote by $\mathbf{p}_t \in \mathbb{R}^n$ the vector of nodal active power injections (and similarly $\mathbf{q}_t$ for reactive injections) when we explicitly refer to injection forecasts.

3.2. Power-flow and baseline state prediction

A standard approach to obtain the electrical state from nodal injections is to solve the AC power-flow equations. At each bus i the net complex injection is $s_i = p_i + jq_i$. The mismatch function for bus i is

$$f_{p,i}(\mathbf{x}) = p_i - |V_i| \sum_{k=1}^{n} |V_k| \times \left(G_{ik}\cos(\phi_i - \phi_k) + B_{ik}\sin(\phi_i - \phi_k)\right), \tag{2}$$

$$f_{q,i}(\mathbf{x}) = q_i - |V_i| \sum_{k=1}^{n} |V_k| \times \left(G_{ik}\sin(\phi_i - \phi_k) - B_{ik}\cos(\phi_i - \phi_k)\right), \tag{3}$$

where $G_{ik} + jB_{ik}$ are the elements of the network admittance matrix $\mathcal{Y}$ [29]. Collecting all mismatches into $\mathbf{f}(\mathbf{x})$, the Newton–Raphson iteration updates the state via

$$\mathbf{x}^{(k+1)} = \mathbf{x}^{(k)} - \mathbf{J}^{-1}(\mathbf{x}^{(k)})\,\mathbf{f}(\mathbf{x}^{(k)}), \tag{4}$$

with $\mathbf{J} = \partial \mathbf{f}/\partial \mathbf{x}$ the Jacobian. Convergence is declared when $\|\mathbf{f}(\mathbf{x})\|$ falls below a tolerance. The AC power flow is solved using a full Newton–Raphson method in polar coordinates, with the forecasted injections treated as fixed inputs.

We use this classical power-flow solver (PF) in two ways:

1) **SLP + PF**: Future nodal injections are generated using standard load profiles (SLPs) and a PV production profile (e.g., a normalized irradiance curve scaled by installed PV capacity). These injections are then used as inputs to an AC power-flow solver to obtain the forecasted electrical state $\mathbf{x}_{t+\tau}$. This pipeline will be used as baseline reference.
2) **Transformer + PF**: Predict future active/reactive power injections at each node using a learned time-series model (Transformer see below) and compute $\mathbf{x}_{t+\tau}$ by solving the power-flow with these predicted injections.

3.3. Self-attention-based transformer

We adopt a self-attention-based Transformer architecture for both (a) multi-horizon nodal injection forecasting (active and reactive powers $\mathbf{p}_t, \mathbf{q}_t$) and (b) direct multi-horizon state forecasting (voltage magnitudes and angles $\mathbf{x}_t$). Let $\mathbf{z}_{t-T_p+1:t}$ denote the past-window measurements. For injection forecasting, this sequence includes historical injections $(\mathbf{p}, \mathbf{q})$ and exogenous covariates $\mathbf{ex}_t$, while for state forecasting it additionally incorporates all available past state measurements. These are combined into node-level feature vectors, concatenated, and linearly projected into a d-dimensional latent space, with positional encodings added to preserve temporal order. The projected sequence is processed by stacked Transformer encoder layers using multi-head self-attention:

$$\text{Attention}(Q, K, V) = \text{softmax}\Big(\frac{QK^\top}{\sqrt{d_k}}\Big)V, \tag{5}$$

where Q, K, V are queries, keys, and values and d_k is the key dimensionality [30]. The encoder yields a hidden representation per input step; we take the final hidden state (time t) as a summary vector and map it with a linear layer to predict the full T_f-step horizon in parallel. Hence the model produces $\hat{\mathbf{x}}_{t+1:t+T_f}$ or $\hat{\mathbf{p}}_{t+1:t+T_f}, \hat{\mathbf{q}}_{t+1:t+T_f}$ in a single forward pass, avoiding iterative autoregression. Transformers are well suited for multi-horizon forecasting because self-attention captures long-range dependencies and flexibly focuses on relevant past observations. Additionally, a LSTM (Long Short-Term Memory) model is included as an additional temporal baseline.

3.4. GNNs

GNNs encode the electrical network as a graph and inject topological inductive biases into node-wise predictions. At each layer a node aggregates neighbor information and updates its state. For a Graph Attention Network (GAT) the update for node i at layer l reads [31]:

$$\mathbf{y}_i^{(l+1)} = \Phi\left(\mathbf{W}_l \mathbf{y}_i^{(l)} + \sum_{j \in \mathcal{N}(i)} \beta_{ij}\, \mathbf{W}_l \mathbf{y}_j^{(l)}\right), \tag{6}$$

where $\mathbf{y}_i^{(l)}$ is the node features, $\mathbf{W}_l$ a learnable matrix, $\mathcal{N}(i)$ the neighbor set, β_{ij} an attention weight, and $\Phi(\cdot)$ a nonlinear activation function. GNNs naturally incorporate node-specific (e.g., voltages, injections) and edge-specific (e.g., impedances, transformer parameters, limits) information, enabling physically informed and topology-aware node-level outputs suitable for state estimation.

3.5. Spatio-temporal integration patterns

To capture spatial and temporal dependencies jointly we evaluate two integration patterns.

1) **Temporal-then-spatial** (Transformer + GAT): A Transformer first produces multi-horizon nodal injection forecasts. These predicted injections are then fed into the GAT from Section 3.4, which maps them to state estimates in a spatially-aware, stepwise fashion.
2) **Spatial-then-temporal**: At each historical time step a GNN converts node features into spatial embeddings; the sequence of per-step embeddings is then processed by a temporal model to produce multi-horizon forecasts for each node.

We implement and compare both strategies. As a spatio-temporal model we use the state–of–the–art A3T-GCN model, which was build for traffic flow prediction and is here applied to distribution system state forecasting for the first time [22].

3.6. A3T-GCN

The A3T-GCN backbone fuses three compact components: a local spatial filter using a Graph Convolutional Network (GCN), a short-term recurrent temporal module based on a Gated Recurrent Unit (GRU), and a temporal-attention aggregation step. Below, we present the core equations and describe the function of each component.

Spatial block. Using the symmetrically-normalized adjacency $\tilde{\mathbf{A}} = \mathbf{D}^{-1/2}(\mathbf{A} + \mathbf{I})\mathbf{D}^{-1/2}$, each time-step input $\mathbf{Y}_t \in \mathbb{R}^{n \times d_{\text{in}}}$ is projected to a spatial embedding

$$S_t = \sigma\big(\tilde{\mathbf{A}}\, \mathbf{Y}_t W_s\big) \in \mathbb{R}^{n \times d_s}, \tag{7}$$

This GCN step injects the grid topology as an inductive bias (see Section 3.4), producing locality-aware node features that make downstream predictions physically consistent and sample-efficient.

Temporal block. Over a look-back window of length T_p, the sequence $\{S_{t-T_p+1}, \dots, S_t\}$ is processed node-wise by a stack of GRUs. For each node i and time index τ we obtain a hidden state $\mathbf{h}_{i,\tau} \in \mathbb{R}^{d_h}$. The GRU captures short-term temporal dependencies while keeping the recurrent state compact and computationally efficient compared to deeper sequence encoders.

Temporal attention. To adaptively focus on relevant past steps, an attention layer reweights the hidden states of each node. For node i,

$$\mathbf{e}_i = \mathbf{v}^\top \tanh\big(\mathbf{H}_i W_a + b_a\big), \tag{8}$$

$$\boldsymbol{\alpha}_i = \text{softmax}(\mathbf{e}_i), \tag{9}$$

$$\mathbf{c}_i = \sum_{\tau=1}^{T_p} \alpha_{i,\tau} \mathbf{h}_{i,\tau}, \tag{10}$$

where $W_a \in \mathbb{R}^{d_h \times d_a}$ and $b_a \in \mathbb{R}^{d_a}$ are learnable parameters, $\mathbf{v} \in \mathbb{R}^{d_a}$ is an attention vector, $\mathbf{e}_i \in \mathbb{R}^{T_p}$ contains unnormalized attention scores, and $\boldsymbol{\alpha}_i$ is the normalized distribution over past steps. The resulting context $\mathbf{c}_i \in \mathbb{R}^{d_h}$ summarizes the temporal history with interpretable importance weights per lag.

Outputs. For multi-horizon prediction, the node contexts $\mathbf{c}_i$ are concatenated with time or horizon embeddings and passed through a linear projection to produce state forecasts $\hat{\mathbf{x}}_{t+1:t+T_f}$ where T_f is the forecast horizon.

In summary: (i) the GCN enforces spatial consistency, (ii) the GRU captures short-term dynamics, and (iii) the attention mechanism selects informative lags, yielding accurate and interpretable node-level forecasts.

Fig. 1 provides a compact overview of all evaluated forecasting pipelines and illustrates how temporal and spatial modeling components are combined in each approach.

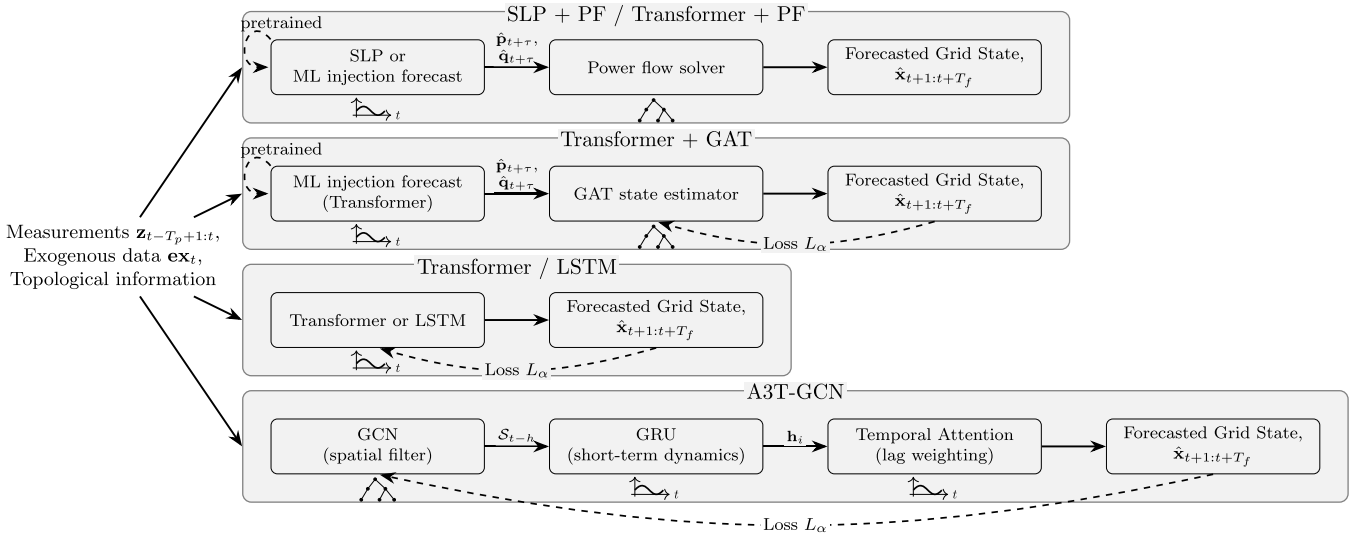


Fig. 1. Flowchart overview of the evaluated forecasting pipelines (left-to-right execution). Measurements, exogenous inputs, and topology information serve as inputs to each model. Rows 1–2 first predict nodal injections and subsequently compute grid states via a power flow solver or a graph-based state estimator. Row 3 performs direct temporal state forecasting without an explicit spatial operator. Row 4 (A3T-GCN) follows an explicit spatial–temporal sequence, where graph convolution (topology-aware spatial filtering) precedes recurrent and attention-based temporal modeling before state decoding. Solid arrows denote inference flow; dashed arrows indicate training feedback via the pinball loss L_α in (11).

3.7. Probabilistic forecasts

For safety-critical operation we prefer probabilistic forecasts that quantify uncertainty. Models are trained to predict quantiles using the pinball (quantile) loss at level $\alpha \in (0, 1)$:

$$L_\alpha(x, \hat{x}) = \begin{cases} \alpha\,(x - \hat{x}) & \text{if } x \geq \hat{x}, \\ (\alpha - 1)\,(x - \hat{x}) & \text{if } x < \hat{x}. \end{cases} \tag{11}$$

Multiple quantile levels are trained jointly within a single model, with the total loss defined as the average pinball loss across quantiles, nodes, state components, and forecast horizons. The pinball loss provides a simple and widely used objective for quantile regression; however, it also has known limitations. In particular, it does not impose consistency across quantiles and can be sensitive to extreme outliers, especially for tail quantiles. Despite these limitations, it offers a robust and computationally efficient way to learn marginal uncertainty estimates, which is well suited to the large-scale and multi-horizon setting considered here. Probabilistic forecasts are produced both for nodal injections and for states: each injection quantile $\hat{\mathbf{p}}_t^\alpha, \hat{\mathbf{q}}_t^\alpha$ can be propagated through the power-flow solver to obtain a corresponding state. Although these propagated states do not represent exact joint quantiles of the full system state, this approach yields practical, scenario-consistent probabilistic state.

3.8. Critical states and operational criteria

To assess operational security, we explicitly evaluate forecasted and true system states against predefined voltage and thermal limits. We consider two types of critical states: *voltage violations* and *line overloadings*. Let $V_{\text{nom}} = 1$ p.u. denote the nominal voltage magnitude. A tolerance ε (here $\varepsilon = 0.05$, corresponding to a $\pm 5\%$ admissible voltage band) defines acceptable voltage deviations, and η (here $\eta = 0.8$) specifies the fraction of the thermal rating used as overload threshold. The corresponding indicators are defined as

$$\mathbb{I}^V_{i,t} = \mathbf{1}\big(|V_{i,t}| < (1-\varepsilon)V_{\text{nom}} \;\vee\; |V_{i,t}| > (1+\varepsilon)V_{\text{nom}}\big), \tag{12}$$

$$\mathbb{I}^L_{ij,t} = \mathbf{1}\Big(|s_{ij,t}| > \eta\, S^{\max}_{ij}\Big), \tag{13}$$

where $\mathbf{1}(\cdot)$ denotes the indicator function and $S^{\max}_{ij}$ the thermal rating of line (i, j). The predicted indicators $\hat{\mathbb{I}}^V_{i,t}$ and $\hat{\mathbb{I}}^L_{ij,t}$ are obtained by applying the same criteria to the forecasted quantities. Concretely, at each time step t the model provides a predicted state $\hat{\mathbf{x}}_t$, from which voltages and line flows are computed (directly or via power-flow calculations based on forecasted injections), and the operational limits are then evaluated on these forecasted values.

4. Results

4.1. Experimental setup

We evaluate the proposed models on two collections of radial low-voltage distribution grids with comparable voltage levels and operational constraints, but differing in realism and controllability. The SimBench cases comprise fully specified networks with synthetic operating scenarios, whereas the Norwegian datasets reflect real-world consumption and generation patterns. This combination enables assessment under controlled benchmark conditions as well as validation on practical low-voltage grids encountered by DSOs.

4.1.1. SimBench low-voltage grids

We use the `SimBench` benchmark [32] via the DSML framework [8], extended to produce `PyTorch Geometric` objects for sequential forecasting. Profiles are at 15-min resolution and include the predefined scenarios "Today", "Tomorrow", and "Day-after-tomorrow". We focus on the "Day-after-tomorrow" scenario because it exhibits the most diverse load patterns and critical states (e.g., EV charging, battery storage, heat pumps). SLP labels from `SimBench` are used to build the profile-based baseline for loads and PV units. To avoid temporal bias, starting points and length are randomly sampled with a minimum series length of 30 days. For state-estimation experiments we use perturbed power-flow outputs as synthetic measurements and compare predictions to noise-free power-flow solutions. We also vary the measurement penetration rate to study observability effects.

4.1.2. Norwegian low-voltage grids

To complement synthetic cases, we evaluate on real Norwegian grids from [33] (185 topologies, 6,809 consumers/producers). Experiments are restricted to nine grids with at least 20 loads, which are explicitly specified in our GitLab repository [34] to ensure reproducibility. Hourly measurements cover January 2022–September 2024. To avoid temporal bias, the time series segments analyzed were selected randomly, with a minimum length of 100 days. In the absence of SLP types we assume residential H0 profiles for loads and treat generation units as PV. Historical exogenous covariates (weather, prices) are used only as past inputs within the lookback window to avoid leakage.

4.1.3. Feature construction

Graph inputs contain node, edge, and structural features:

- **Node features:** historical measurements (voltage magnitude, active/reactive power), bus attributes (voltage level, slack/zero-injection flags), temporal indicators (time index, day-of-week, hour-of-day), and—where available—historical exogenous covariates (weather, price).
- **Edge features:** measured power flows (only at transformers), line/transformer parameters (conductance, susceptance, shunts), and operational attributes (limits, transformer ratings, phase shifts, on/off status).
- **Graph structure:** bus connectivity via lines or transformers.
- **Targets:** voltage magnitudes and angles from power-flow solutions.

4.1.4. Implementation

All models are implemented in `PyTorch`, using `PyTorch Geometric` / `PyTorch Geometric Temporal` and `PyTorch Lightning`. Power flows are solved with `pandapower` [35]. Power-flow convergence was monitored using the standard Newton–Raphson stopping criterion on the mismatch norm. Forecast instances for which the solver did not converge within the default iteration limit were rare and excluded from evaluation to ensure comparability across models. Training uses a rolling-window scheme with per-node normalization, early stopping on a validation split, dropout, and the Adam optimizer. Hyperparameters were briefly tuned and then fixed across models unless stated, with a hidden dimension of 32 and dropout of 0.3. The Transformer uses 4 attention heads and, both Transformer and LSTM consists of two stacked layers plus a linear output. We adopt an 0.8/0.1/0.1 train/validation/test split. During testing the rolling window advances by T_f (non-overlapping), so each test time point is forecast exactly once. The full codebase and selected grid specifications are provided in our GitLab for reproducibility [34].

4.1.5. Evaluation metrics

We analyze deterministic and probabilistic metrics separately for voltage magnitudes and angles.

For the median forecast we compute mean absolute error (MAE)

$$\mathrm{MAE}(\mathbf{x}, \hat{\mathbf{x}}) = \frac{1}{n} \sum_{i=1}^{n} |x_i - \hat{x}_i|, \tag{14}$$

with $\mathbf{x}$ and $\hat{\mathbf{x}}$ denoting ground truth and forecast across n buses. Results are averaged across buses and forecast horizons.

To assess the quality of full predictive distributions, we use the Continuous Ranked Probability Score (CRPS), which measures the overall calibration and sharpness of probabilistic forecasts. For a predictive cumulative distribution function (CDF) F and an observation x, the CRPS is defined as [36]:

$$\mathrm{CRPS}(F, x) = \int_{-\infty}^{\infty} \left(F(y) - \mathbf{1}\{y \geq x\} \right)^2 dy, \tag{15}$$

and approximate it numerically from model quantile outputs via the integral of pinball losses over quantile levels using the trapezoidal rule. Lower CRPS indicates sharper and better-calibrated predictive distributions.

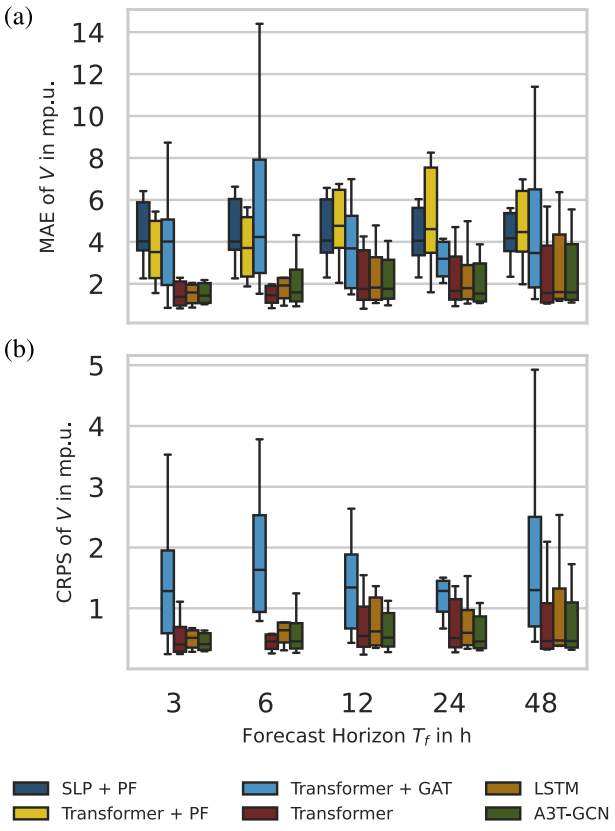


Fig. 2. Box plots of voltage magnitude V forecast metrics for `SimBench` low voltage grids: (a) MAE and (b) CRPS, across varying forecast horizons T_f with a fixed lookback window of $T_p = 24$h. Note that in the CRPS plot, the Transformer + PF box is cropped because its propagated quantiles are not directly comparable to the state-forecast quantiles and produce much larger values.

For voltage violations and line overloads we report binary classification metrics:

$$\text{Precision} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FP}}, \qquad \text{Recall} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FN}}, \tag{16}$$

where TP, FP, FN are true positives, false positives, and false negatives. Precision reflects the proportion of predicted violations that are correct, while recall quantifies the fraction of actual violations that are successfully identified.

4.2. Evaluation

4.2.1. SimBench low voltage grids

We evaluated all forecasting methods across forecast horizons T_f with a fixed lookback $T_p = 24$h. For each grid, test-set metric scores were computed and each score contributes a single point in the box plots. Because voltage-angle forecasts exhibit the same trends, we omit them and focus on voltage magnitudes. Fig. 2(a) shows that MAE generally increases with T_f, reflecting the greater uncertainty of longer horizons; the SLP + PF is the sole exception, remaining approximately constant because it always predicts fixed standard load profiles. The SLP + PF's variance results from mismatch between predefined profiles and actual grid behavior, an effect amplified in the examined `SimBench` "Day-after-tomorrow" scenario where EV charging, storage and heat-pump patterns diversify loads.

Among data-driven methods, A3T-GCN attains the lowest median MAE and the smallest variance across most horizons, demonstrating accuracy and robustness. The direct state-forecasting Transformer is the next strongest, followed by the LSTM; both, however, show greater variability. Injection-forecasting pipelines are less stable overall: applying a

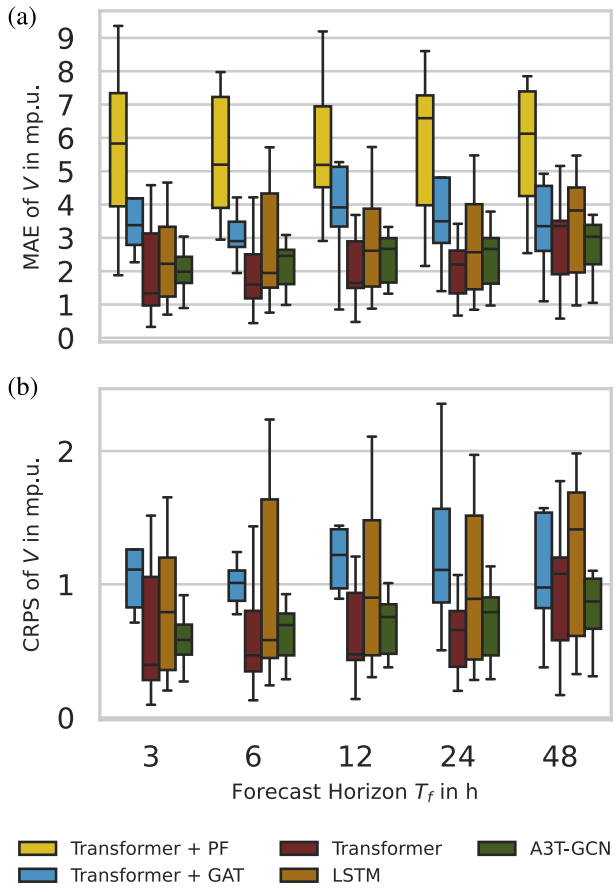


Fig. 3. Box plots of voltage magnitude V forecast metrics for low-voltage grids in Norway: (a) MAE and (b) CRPS, across varying forecast horizons T_f with a fixed lookback window of $T_p = 24$h. Note that in (a) the SLP + PF box is cropped due to extreme MAE values from mismatched profiles; in (b) the Transformer + PF box is cropped because its propagated quantiles are not directly comparable to the state-forecast quantiles and produce much larger CRPS values.

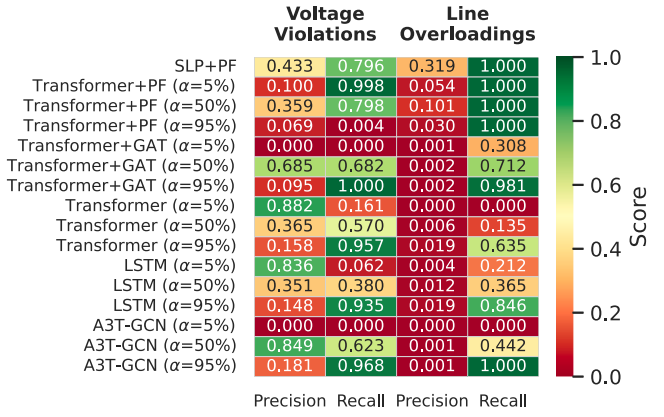


Fig. 4. The heatmap shows precision and recall for voltage limit violations (left) and line overloads (right) across all evaluated models for a forecast horizon of $T_f = 24$h. The analysis focuses on the "Day-after-tomorrow" `SimBench` scenario, where critical events as defined in Section 3.8 occur.

power-flow solver to predicted injections compounds errors, and the Transformer + PF pipeline performs worst because the injection forecaster is not trained for state accuracy. Adding a GAT post-processor can improve results, but the two-stage approach, requiring separate training for forecasting and state mapping, remains more variable and degrades faster with increasing T_f. These observations suggest that jointly modeling spatial and temporal dependencies (as in A3T-GCN) is more effective.

Similar trends appear in the probabilistic evaluation (Fig. 2(b)). The CRPS is lowest for the A3T-GCN, which also shows minimal variance, indicating accurate and well-calibrated uncertainty estimates. The Transformer and LSTM remain competitive but more variable, while injection-based pipelines lag behind. The SLP + PF model is omitted since it lacks predictive distributions, and Transformer + PF is excluded because its quantiles are not directly comparable. Overall, these results confirm that integrating spatial and temporal dependencies yields the most reliable deterministic and probabilistic forecasts.

4.2.2. Low-voltage grids from norway

The evaluation on the Norwegian low-voltage grid datasets confirms the generalizability of the proposed approach. Overall trends mirror those observed for the `SimBench` cases (see Fig. 3(a) and (b)), supporting the robustness of the A3T-GCN across different network topologies and data sources, as it exhibits the lowest variance across forecast horizons T_f. While the Transformer outperforms the A3T-GCN on individual grids, likely because the hourly resolution makes temporal patterns easier to capture, its performance is less consistent across networks. This adaptability of the A3T-GCN to varying grid structures highlights its potential as a more generalizable model that could be applied to previously unseen systems.

Predefined SLPs capture Norwegian load and generation behavior less accurately than in `SimBench`, and their performance therefore degrades notably: the median MAE for voltage-magnitude forecasts exceeds 30mp.u., underscoring the importance of accurate, grid-specific profiles for profile-based baselines.

4.2.3. Critical state detection

To further evaluate how well the models capture critical grid conditions as described in Section 3.8, we analyzed test periods with voltage violations or line overloads in the "Day-after-tomorrow" `SimBench` scenario. As results vary little across forecast horizons, we present $T_f = 24$h. Fig. 4 shows heatmaps indicating whether each method correctly predicted a voltage or loading violation at each bus or line over time.

For DSOs, *recall*, detecting all actual violations, is crucial. The upper quantiles ($\alpha = 95\%$) of probabilistic models and the lower quantile ($\alpha = 5\%$) of Transformer + PF achieve the highest recall, consistent with predominantly overvoltage events from high PV generation. The SLP + PF also captures many violations due to PV inclusion in profiles. Higher recall comes at the cost of *precision*, whereas median ($\alpha = 50\%$) forecasts provide a more balanced trade-off, particularly the A3T-GCN, which maintains high values for both metrics. Using multiple forecast quantiles thus helps DSOs assess the likelihood and severity of upcoming violations for risk-aware operational decisions.

Line overloads remain harder to predict: state-forecasting models show either low precision or recall, and predicted quantiles poorly match true load distributions, though the $\alpha = 95\%$ quantile attains the highest recall. These results suggest that dedicated line-loading models are likely more suitable for capturing such events.

4.2.4. Measurement penetration rate

To further analyze the sensitivity of the forecasting models to data availability, we conducted an additional study on the `SimBench` grids, varying the measurement penetration rate. In this experiment, the rate was reduced to 50%, meaning that only half of the buses with load or generation were equipped with measurements. For unmeasured buses, the provided profiles were used as pseudo-measurements. The SLP + PF model is not affected by this reduction of measurements.

The results reveal mixed effects, with average values summarized in Table 1. For the purely temporal models, performance changes were minor and inconsistent, showing both slight improvements and degradations depending on the grid. Similar variability was observed for the

Table 1
Mean forecasting performance for different measurement rates and methods.

Method	MAE VM in mp.u.		CRPS VM in mp.u.	
	50	100	50	100
	Measurement penetration rate in %			
Transformer + PF	3.54	3.69	6.07	5.90
Transformer + GAT	3.50	2.69	1.20	1.30
LSTM	1.46	1.36	0.47	0.44
Transformer	1.32	1.38	0.40	0.44
A3T-GCN	1.52	1.37	0.44	0.39

less stable two-stage pipelines involving the Transformer-based injection forecasts, suggesting that these models rely primarily on temporal patterns and that the influence of individual measurements is limited. In contrast, the A3T-GCN consistently benefited from higher measurement availability across all four tested grids, highlighting both its robustness and its dependence on accurate measurement data. This trend suggests that the effect may become even more pronounced at lower penetration levels, potentially posing challenges for low-voltage grids with limited measurement availability.

5. Discussion

The results demonstrate that the proposed A3T-GCN effectively fuses spatial and temporal dependencies for accurate state forecasting, and consistently matches or improves upon purely temporal models such as Transformers and LSTMs across most settings. At the same time, the results also indicate that strong temporal baselines can achieve competitive pointwise accuracy when sufficient historical data are available, even under reduced measurement penetration. This highlights that temporal correlations alone already explain a large fraction of the variance in nodal states, particularly in stable operating regimes.

A key assumption of graph-based models is the availability of accurate grid topology information. In practice, however, topology data may be incomplete, outdated, or erroneous, which can degrade performance since spatial message passing directly depends on the encoded connectivity. At the same time, explicitly incorporating topology can be a structural strength. By embedding electrical connectivity into the architecture, models such as A3T-GCN introduce a principled inductive bias reflecting physical coupling between nodes. This structured representation enables the exploitation of spatial correlations and may support more systematic adaptation when operating conditions shift or when transferring models across grids with different layouts. In contrast, purely temporal models absorb spatial effects implicitly, which may limit generalization under structural changes. Graph-based formulations therefore offer a promising pathway toward scalable and transferable learning across heterogeneous distribution networks. Robustness to topology uncertainty and cross-network generalization remain important directions for future research, including evaluation within modular environments such as the DSML framework [8].

Predicting critical operating conditions, particularly line or transformer overloads, remains challenging when relying solely on nodal state forecasts. This underscores the importance of dedicated models that explicitly account for line-loading dynamics. Furthermore, higher temporal resolution could reveal short-lived or transient grid states that are obscured in hourly or quarter-hourly data.

6. Conclusion

This paper studied data-driven approaches for probabilistic short- to mid-term state forecasting in low-voltage distribution networks. We compared physics-based baselines, purely temporal models, and spatio-temporal graph-based architectures across synthetic benchmark grids and real-world networks, using both forecast accuracy and operationally relevant evaluation criteria. The results show that while temporal models capture a large share of system dynamics, incorporating network topology improves robustness and structural consistency across heterogeneous grids. The proposed A3T-GCN achieved the most stable forecasts overall, demonstrating the benefit of explicitly modeling electrical connectivity alongside temporal patterns.

Beyond predictive performance, the results highlight the practical potential of probabilistic state forecasting for distribution system operators. Quantile forecasts enable risk-aware operational planning by identifying time periods with elevated probabilities of voltage violations or line overloads. Such risk assessments can, for example, inform the derivation of dynamic operating envelopes, i.e., time-varying flexibility limits for distributed energy resources based on anticipated grid conditions [37]. In this way, forecasting models can become an integral component of digital DSO toolchains, supporting proactive congestion management and renewable integration.

Future work will address cross-network generalization, robustness to topology uncertainty, higher temporal resolution, the exploration of fully distributional probabilistic forecasting approaches, and the systematic integration of forecast-based decision support into closed-loop operational workflows. Overall, the findings indicate that spatio-temporal learning architectures constitute a promising building block for scalable, data-driven grid operation in increasingly complex distribution systems.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process During the preparation of this work the author(s) used ChatGPT in order to improve the readability of the text. After using this tool, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the published article.

CRediT authorship contribution statement

Theodor Haug: Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization; **Christian Rehtanz:** Writing – review & editing; **Christoph Goebel:** Writing – review & editing, Supervision, Conceptualization.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Theodor Haug reports financial support was provided by Federal Ministry for Economic Affairs and Energy. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

All data used in this study are open source and publicly available. Details and links to the repositories are provided in the manuscript.

References

[1] I.E.A. (IEA), Executive summary – unlocking the potential of distributed energy resources – analysis, 2022, https://www.iea.org/reports/world-energy-outlook-2023/executive-summary. Accessed: 2025-10-13.
[2] C. Rudin, D. Waltz, R.N. Anderson, A. Boulanger, A. Salleb-Aouissi, M. Chow, H. Dutta, P.N. Gross, B. Huang, S. Ierome, D.F. Isaac, A. Kressner, R.J. Passonneau, A. Radeva, L. Wu, et al., Machine learning for the New York city power grid, IEEE Trans. Pattern Anal. Mach. Intell. 34 (2) (2012) 328–345. Conference Name: IEEE Transactions on Pattern Analysis and Machine Intelligence, https://doi.org/10.1109/TPAMI.2011.108
[3] A.E.B. Abu-Elanien, M.M.A. Salama, K.B. Shaban, Modern network reconfiguration techniques for service restoration in distribution systems: a step to a smarter grid, Alex. Eng. J. 57 (4) (2018) 3959–3967. https://doi.org/10.1016/j.aej.2018.03.011
[4] A. Abur, A.G. Expósito, Power System State Estimation: Theory and Implementation, CRC Press, 2004.

[5] K. Dehghanpour, Z. Wang, J. Wang, Y. Yuan, F. Bu, et al., A survey on state estimation techniques and challenges in smart distribution systems, IEEE Trans. Smart Grid 10 (2) (2019) 2312–2322. https://doi.org/10.1109/TSG.2018.2870600
[6] K.R. Mestav, J. Luengo-Rozas, L. Tong, et al., Bayesian state estimation for unobservable distribution systems via deep learning, IEEE Trans. Power Syst. 34 (6) (2019) 4910–4920. Conference Name: IEEE Transactions on Power Systems, https://doi.org/10.1109/TPWRS.2019.2919157
[7] A.S. Zamzam, N.D. Sidiropoulos, Physics-aware neural networks for distribution system state estimation, IEEE Trans. Power Syst. 35 (6) (2020) 4347–4356. Conference Name: IEEE Transactions on Power Systems, https://doi.org/10.1109/TPWRS.2020.2988352
[8] T. Haug, C. Goebel, DSML: A comprehensive framework for evaluating ML methods for distribution system state estimation, in: Proceedings of the 16th ACM International Conference on Future and Sustainable Energy Systems, E-Energy '25, Association for Computing Machinery, New York, NY, USA, 2025, pp. 122–135. https://doi.org/10.1145/3679240.3734629
[9] B.P. Hayes, M. Prodanovic, State forecasting and operational planning for distribution network energy management systems, IEEE Trans. Smart Grid 7 (2) (2016) 1002–1011. Conference Name: IEEE Transactions on Smart Grid, https://doi.org/10.1109/TSG.2015.2489700
[10] R. Dobbe, D. Arnold, S. Liu, D. Callaway, C. Tomlin, Real-time distribution grid state estimation with limited sensors and load forecasting, in: 2016 ACM/IEEE 7th International Conference on Cyber-Physical Systems (ICCPS), 2016, pp. 1–10. https://doi.org/10.1109/ICCPS.2016.7479117
[11] H. Yue, W. Zhang, U.C. Yilmaz, T. Yildiz, H. Huang, H. Liu, Y. Lin, A. Abur, N. University-Boston, et al., Graph-learning-assisted state estimation using sparse heterogeneous measurements Electr. Power Syst. Res. 235 (2024) 110644. https://doi.org/10.1016/j.epsr.2024.110644
[12] L. Zhang, G. Wang, G.B. Giannakis, et al., Real-time power system state estimation and forecasting via deep unrolled neural networks, IEEE Trans. Signal Process. 67 (15) (2019) 4069–4077. https://doi.org/10.1109/TSP.2019.2926023
[13] D. Mukherjee, S. Chakraborty, S. Ghosh, et al., Power system state forecasting using machine learning techniques, Electr. Eng. 104 (1) (2022) 283–305. https://doi.org/10.1007/s00202-021-01328-z
[14] S. Liao, Y. Liu, J. Xu, L. Jia, D. Ke, X. Jiang, Data-driven real-time congestion forecasting and relief with high renewable energy penetration, IEEE Trans. Ind. Inf. 21 (1) (2025) 12–29. https://doi.org/10.1109/TII.2024.3413290
[15] J. Simeunović, B. Schubnel, P.-J. Alet, R.E. Carrillo, Spatio-temporal graph neural networks for multi-site PV power forecasting, IEEE Trans. Sustain. Energy 13 (2) (2022) 1210–1220. https://doi.org/10.1109/TSTE.2021.3125200
[16] Y. Yang, Y. Liu, Y. Zhang, S. Shu, J. Zheng, DEST-GNN: a double-explored spatio-temporal graph neural network for multi-site intra-hour PV power forecasting, Appl. Energy 378 (2025) 124744. https://doi.org/10.1016/j.apenergy.2024.124744
[17] J. Park, J. Park, Physics-induced graph neural network: an application to wind-farm power estimation, Energy 187 (2019) 115883. https://doi.org/10.1016/j.energy.2019.115883
[18] M. Khodayar, J. Wang, Spatio-temporal graph deep neural network for short-term wind speed forecasting, IEEE Trans. Sustain. Energy 10 (2) (2019) 670–681. https://doi.org/10.1109/TSTE.2018.2844102
[19] G. Yu, C. Liu, B. Tang, R. Chen, L. Lu, C. Cui, Y. Hu, L. Shen, S.M. Muyeen, Short term wind power prediction for regional wind farms based on spatial-temporal characteristic distribution, Renew. Energy 199 (2022) 599–612. https://doi.org/10.1016/j.renene.2022.08.142
[20] C. Wei, D. Pi, M. Ping, H. Zhang, Short-term load forecasting using spatial-temporal embedding graph neural network, Electr. Power Syst. Res. 225 (2023) 109873. https://doi.org/10.1016/j.epsr.2023.109873
[21] W. Zhuang, J. Fan, M. Xia, K. Zhu, A multi-scale spatial–temporal graph neural network-based method of multienergy load forecasting in integrated energy system, IEEE Trans. Smart Grid 15 (3) (2024) 2652–2666. https://doi.org/10.1109/TSG.2023.3315750
[22] J. Zhu, Y. Song, L. Zhao, H. Li, et al., A3T-GCN: attention temporal graph convolutional network for traffic forecasting, 2020. arXiv:2006.11583 [cs], https://doi.org/10.48550/arXiv.2006.11583
[23] A. Berezin, S. Balduin, T. Oberließen, E. Veith, S. Peter, S. Lehnhoff, et al., Application of recurrent graph convolutional networks to the neural state estimation problem, Int. J. Electr. Electron. Eng. Telecommun. (2023) 209–215. Publisher: EJournal Publishing, https://doi.org/10.18178/ijeetc.12.3.209-215
[24] H. Wu, Z. Xu, M. Wang, et al., Unrolled spatiotemporal graph convolutional network for distribution system state estimation and forecasting, IEEE Trans. Sustain. Energy 14 (1) (2023) 297–308. https://doi.org/10.1109/TSTE.2022.3211706
[25] J. Lin, M. Tu, H. Hong, C. Lu, W. Song, et al., Spatiotemporal graph convolutional neural network-based forecasting-aided state estimation using synchrophasors, IEEE Internet Things J. 11 (9) (2024) 16171–16183. https://doi.org/10.1109/JIOT.2024.3351850
[26] A. Srivastava, D. Steen, L.A. Tuan, O. Carlson, et al., A congestion forecast framework for distribution systems with high penetration of PVs and PEVs, in: 2019 IEEE Milan PowerTech, IEEE, Milan, Italy, 2019, pp. 1–6. https://doi.org/10.1109/PTC.2019.8810871
[27] M. Antončič, M. Ilkovski, B. Blažič, State forecasting in distribution networks, in: 2019 IEEE PES Innovative Smart Grid Technologies Europe (ISGT-Europe), 2019, pp. 1–5. https://doi.org/10.1109/ISGTEurope.2019.8905521
[28] A. Hernandez-Matheus, K. Berg, V. Gadelha, M. Aragüé-Peñalba, E. Bullich-Massagué, S. Galceran-Arellano, et al., Congestion forecast framework based on probabilistic power flow and machine learning for smart distribution grids, Int. J. Electr. Power Energy Syst. 156 (2024) 109695. https://doi.org/10.1016/j.ijepes.2023.109695
[29] A.R. Bergen, V. Vittal, Power Systems Analysis, Prentice Hall, 2000. Google-Books-ID: 4InAQwAACAAJ.
[30] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A.N. Gomez, U. Kaiser, I. Polosukhin, et al., Attention is all you need, in: Advances in Neural Information Processing Systems, vol. 30, Curran Associates, Inc., 2017. https://arxiv.org/abs/1706.03762.
[31] P. Veličković, G. Cucurull, A. Casanova, A. Romero, P. Liò, Y. Bengio, et al., Graph attention networks, 2018. arXiv:1710.10903 [stat] https://doi.org/10.48550/arXiv.1710.10903
[32] C. Spalthoff, D. Sarajlic, C. Kittl, S. Drauz, T. Kneiske, C. Rehtanz, M. Braun, SimBench: open source time series of power load, storage and generation for the simulation of electrical distribution grids, in: International ETG-Congress 2019; ETG Symposium, 2019, pp. 1–6. https://ieeexplore.ieee.org/abstract/document/8836022/authorsauthors.
[33] J. Maree, Energy distribution models with AMI smart meter sensor dataset 1 (2025). Publisher: Mendeley Data, https://doi.org/10.17632/jv3rz8k35r.1
[34] T. Haug, GitLab repository: DSML: a comprehensive framework for evaluating ML methods for distribution system state estimation, 2025, https://gitlab.lrz.de/energy-management-technologies-public/dsml.
[35] L. Thurner, A. Scheidler, F. Schäfer, J.-H. Menke, J. Dollichon, F. Meier, S. Meinecke, M. Braun, et al., Pandapower—an open-source python tool for convenient modeling, analysis, and optimization of electric power systems, IEEE Trans. Power Syst. 33 (6) (2018) 6510–6521. Conference Name: IEEE Transactions on Power Systems, https://doi.org/10.1109/TPWRS.2018.2829021
[36] T. Gneiting, A.E. Raftery, Strictly proper scoring rules, prediction, and estimation, J. Am. Stat. Assoc. 102 (477) (2007) 359–378. https://doi.org/10.1198/016214506000001437
[37] A. Wickramasinghe, M. Vilathgamuwa, G. Nourbakhsh, P. Corry, et al., A review of dynamic operating envelopes: computation, applications and challenges, Modelling 6 (2) (2025) 29. Number: 2 Publisher: Multidisciplinary Digital Publishing Institute, https://doi.org/10.3390/modelling6020029